



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Converting Units

The method: dimensional analysis

Dimensional analysis is a formal-sounding name for a simple habit: *multiply by fractions that equal one, chosen so the units you do not want cancel.*

A fraction like $\frac{1000 \text{ m}}{1 \text{ km}}$ equals **one**, because 1000 m is 1 km. Multiplying by one never changes the value — only the units it is written in.

So: put the unit you want to **get rid of** on the opposite side of the fraction line, and cross it out. Keep going until only the units you want remain.

Worked example 1 — how many seconds in a day?

Start with what you know, and cancel your way across:

$$\frac{60 \text{ s}}{1 \cancel{\text{min}}} \times \frac{60 \cancel{\text{min}}}{1 \cancel{\text{h}}} \times \frac{24 \cancel{\text{h}}}{1 \text{ day}} = \frac{60 \times 60 \times 24 \text{ s}}{1 \text{ day}} = \frac{86\,400 \text{ s}}{1 \text{ day}}$$

Minutes cancel against minutes, hours against hours. Only seconds and days are left — which is exactly what was asked for. **Then** multiply.

Worked example 2 — 72 km/h into m/s

Two things must change: kilometres into metres, and hours into seconds.

$$\frac{72 \cancel{\text{km}}}{1 \cancel{\text{h}}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} \times \frac{1 \cancel{\text{h}}}{3600 \text{ s}} = \frac{72 \times 1000}{3600} \frac{\text{m}}{\text{s}} = 20 \text{ m/s}$$

Sense check: 72 km/h is motorway-ish, and 20 m/s is about the same. If your answer had come out as 259 200, you multiplied where you should have divided.

Conversion factors you should know

Length

1 km = 1000 m

1 m = 100 cm

1 cm = 10 mm

Time

1 min = 60 s

1 h = 3600 s

1 d = 86 400 s

Mass

1 kg = 1000 g

1 t = 1000 kg

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1) **Convert** each of these. Show the fractions and cross out the units that cancel.

a) 5 km into metres [1]

b) 2.5 h into seconds [2]

c) 450 cm into metres [1]

2) **Convert** these speeds into m/s. This is the one that matters most — every physics equation needs speed in m/s.

a) 36 km/h [3]

b) 108 km/h [3]

3) **Convert** 25 m/s into km/h. You are going the other way now, so the fractions flip. [3]

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- 4) **Calculate** the energy in the kinetic store of a 1400 kg car travelling at 90 km/h. [5]

Two steps: Convert the speed first, then use $E_k = \frac{1}{2}mv^2$. Marks are given for the conversion *and* for the calculation, so show both.

- 5) **Explain** why a physicist would not simply put 90 into the kinetic energy equation, even though 90 is the number written on the speedometer. [2]

- 6) **Determine** whether a cheetah at 104 km/h or a peregrine falcon at 89 m/s is faster. Convert one of them so you are comparing like with like, and say which you chose to convert and why. [4]